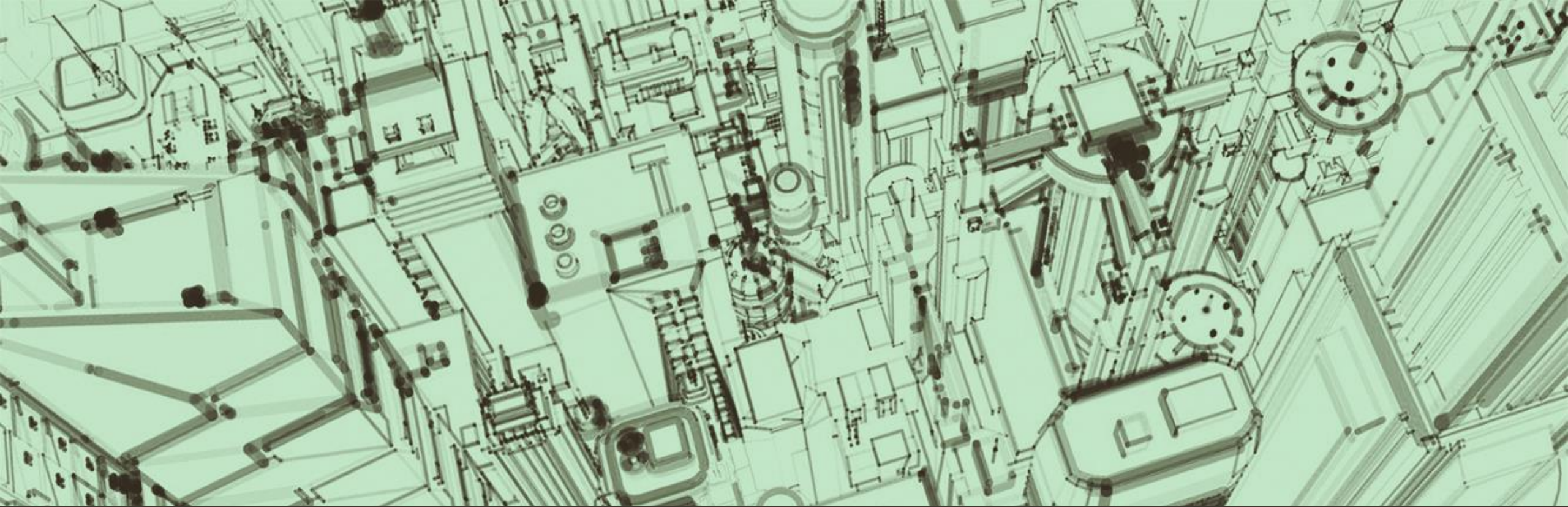




InsureTech

MIT Fintech Lecture Series 2025



What we would cover...

- ✓ Fundamentals of Insurance
- ✓ The Insurance Value Chain
- ✓ Friction points in Insurance
- ✓ InsureTech Solutions
- ✓ InsureTech Organizational Examples

Fundamentals of Insurance

- Concepts
 - Value of Life, Property
 - Uncertainty >> Risk >> Hazard/Peril >> Loss
 - Risk Management Tool >> Transfer of risk from insured to insurer >> Homogenous Pools
 - Contract between the Insurer and the Insured
 - Mathematics of Insurance Pricing >> Probability/Calculus/Statistics/Financial Maths, etc.
- Principles of Insurance
 - Uberrimae Fides (Utmost Good Faith)
 - Insurable Interest
 - Indemnity
 - Contribution
 - Subrogation
 - Loss Minimization
 - Causa Proxima (Proximate Cause)

Fundamentals of Insurance

- Types of Insurance
 - Life Insurance
 - Life (Risk of Living too Short!)
 - Health Assurance
 - Retirement Planning (Risk of Living too Long!)
 - General Insurance
 - Health Insurance
 - Property & Casualty (P&C)
 - Speciality Insurance
- Regulation & Compliance
 - IRDA >> Key Governing Body >> Industry Development, Consumer Protection
 - Formation, Capital Requirements, Products, Pricing, Investments, Grievance Redressal, Ombudsman
 - Other impacting regulations:
 - Companies Act, SEBI, Contracts Act, Data Protection Act, Income Tax Act, Motor Vehicles Act, etc.

Understanding the Needs

- Protection
- Savings & Investment
- Tax Planning
- Estate Planning / Will

Understanding Products

- Life: Pure Term
- Life: Endowment
- Life: Unit Linked
- Life: KeyMan
- Life: Retirement Plans/Pensions – Defined Benefit / Defined Contribution Plans
- General: Health Plans
- General: Motor
- General: Property
- General: Miscellaneous (Travel, Gadgets, Equipment)
- General: Commercial
- General: Speciality
- Group

Understanding Channels

- Agency / Corporate Agency
- Broker / Corporate Broker
- Bancassurance
- Direct Marketing
- Digital
- Embedded
- Re-insurance

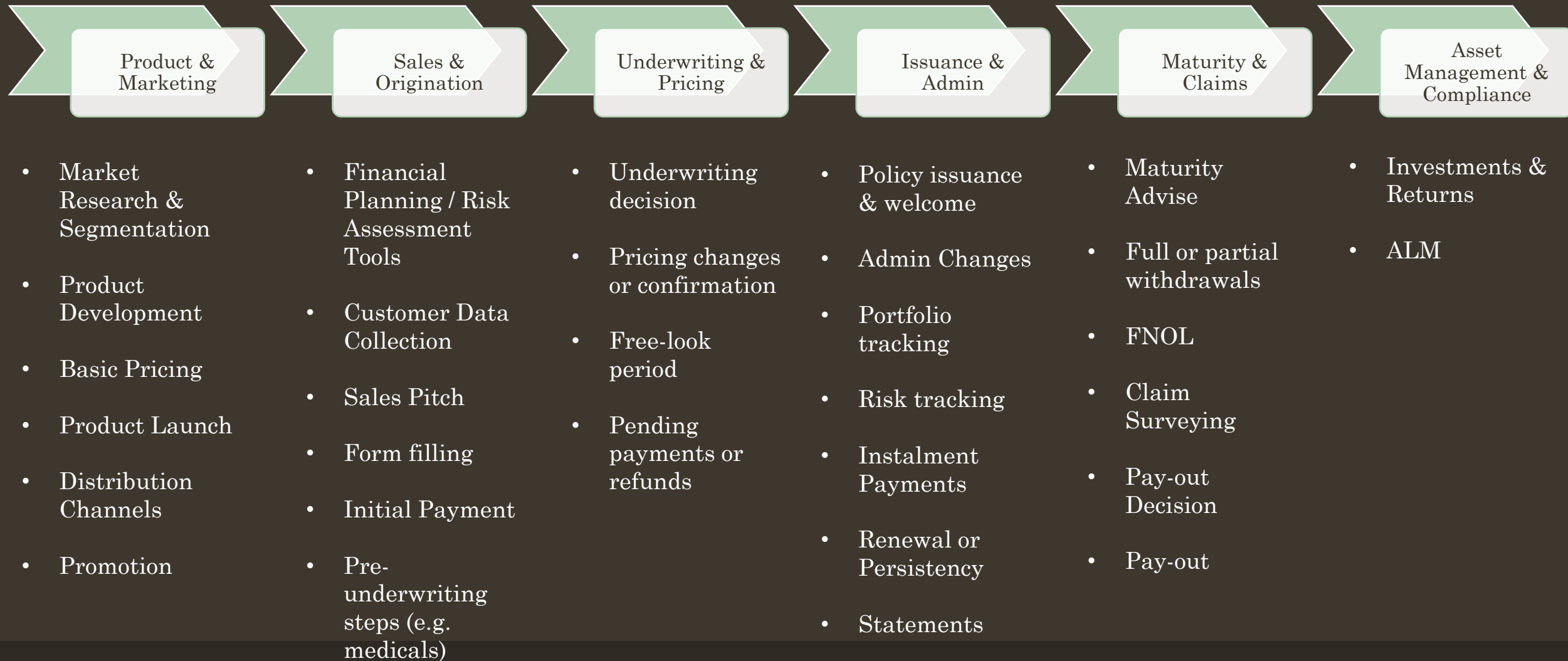
Lets check your knowledge...

- Sum Assured
- Premium
- IDV
- HLV
- PA Cover
- Co-Pay / Deductible
- Underwriting
- Insurance Agent
- Corporate Agent
- Insurance Brokers
- Bancassurance
- Ombudsman
- MRTA
- Top Insurance Companies of the world (Name 5-10)
- Insurance Companies in India (Name 5-10)
- Insurance Brokers (Can you name some?)

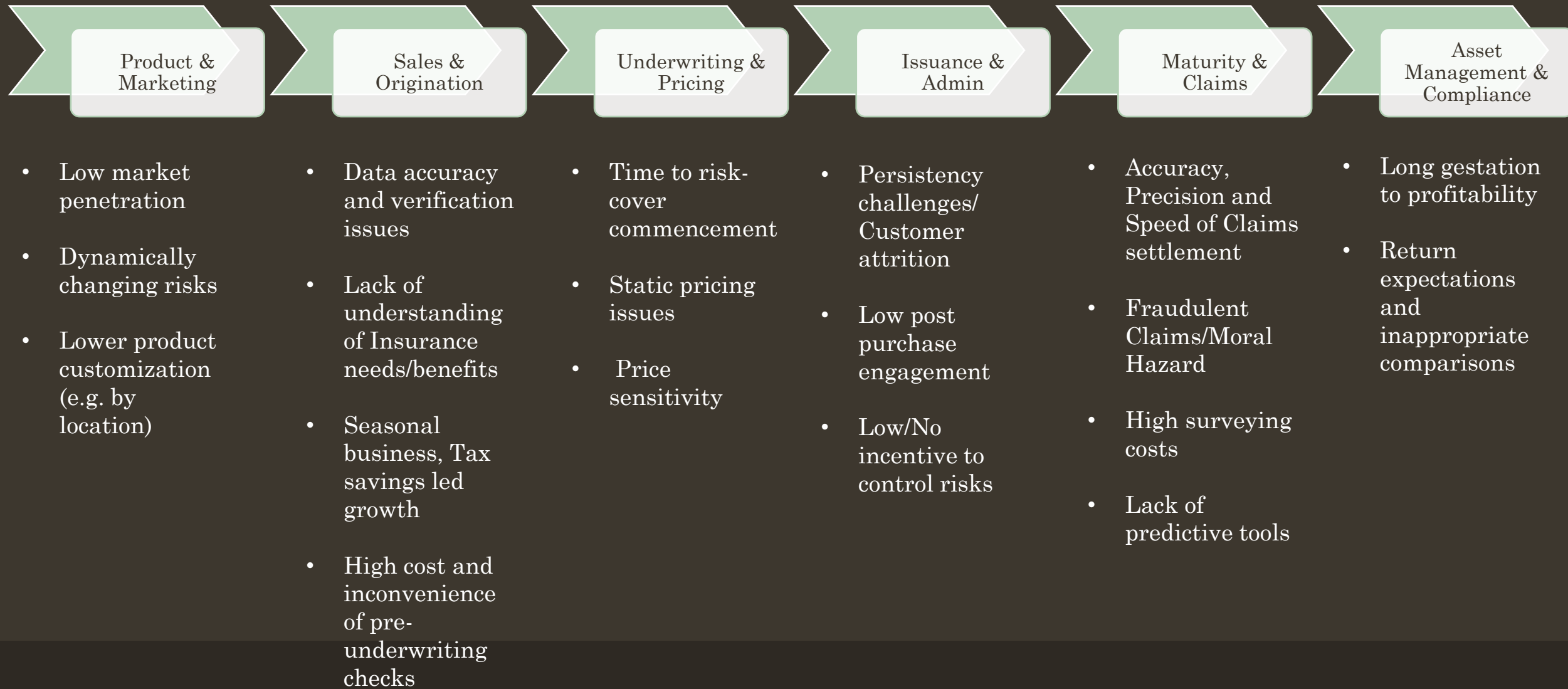
Some Insurers in India...



The Insurance Value Chain



Friction & Pain Points across the Value Chain



Other triggers to InsureTech

- Low market penetration
- Changing customer needs – Speed, Simplicity, Costs, Benefits
- Digital adoption

Tectonic Shifts

- From Managing Policies to Managing Risks
- From Claim Settlement to Claim Prevention
- From Supply Driven models to Demand Driven models

What are real startup upto

- Improving Customer Experience
- Enabling Incumbent Providers
- Risk Management
- Superior Products
- New Markets
- New Business Models

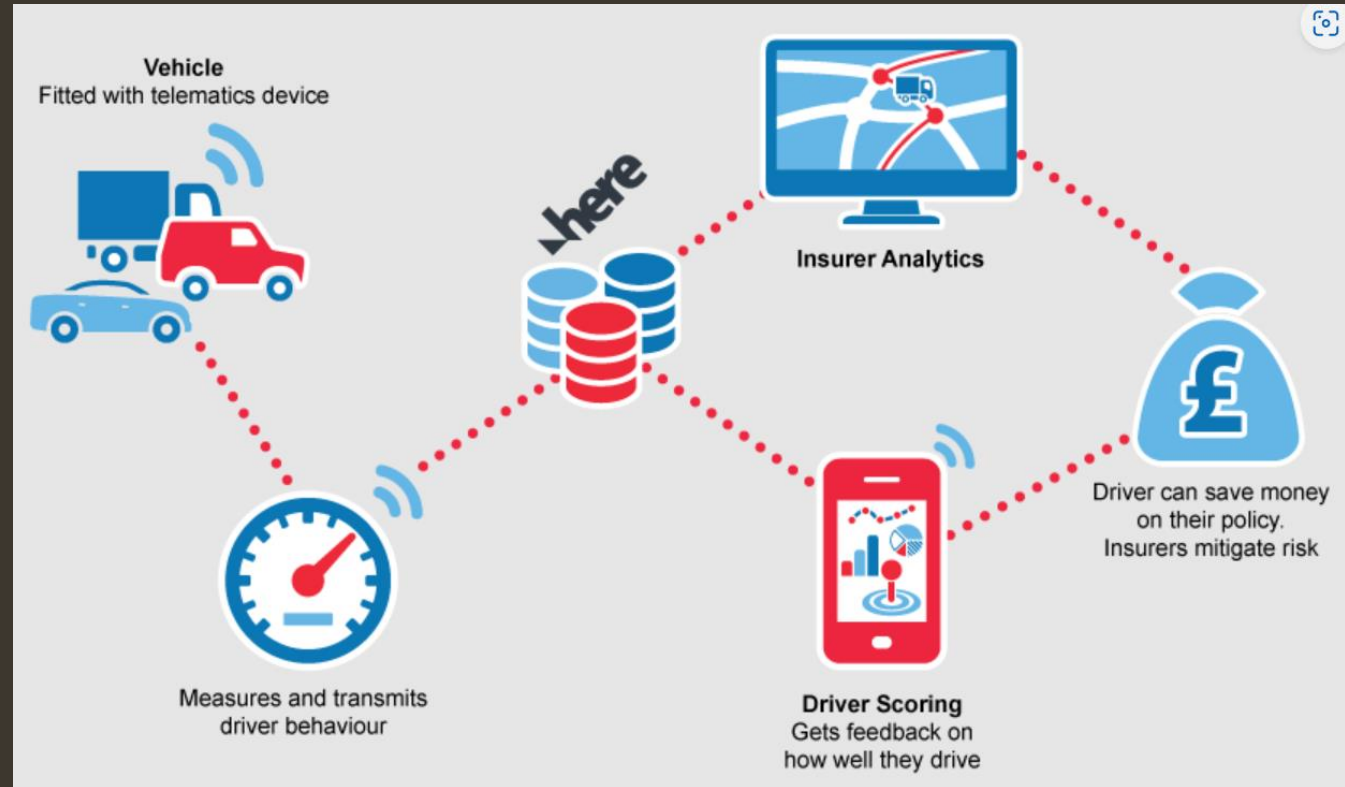
6 Mega-trends

- From Reactive to Proactive
- From Customership to Membership
- From Transactional to Social
- From Approximate to Individual
- From Analogue to Digital
- From Local to Global

InsureTech Solutions / Telematics

Telematics in Motor Insurance

- 'Black Box' of the vehicle
- Record & transmission of driver metrics: location, time of day, mileage, driving frequency, behaviour around hazardous zones, speed, rates of acceleration and braking habits.
- Opportunity to improve driving behaviour
- Personalized Pricing



InsureTech Solutions / Wearables

Wearables in Health Insurance

- Smart watches, clothes, shoes, etc.
- Records & transmission of biometric information
 - Physical activity
 - Cardiovascular activity
 - Sleep data
 - BP and Sugar
 - Pollution exposure
- Opportunity to focus on wellness
- Personalized Pricing



The Future Is Now: Wearables
for Insurance Risk Assessment

InsureTech Solutions / IoT @ Home

IoT Devices in Home/Office Insurance

- Home Security
- Flood, Smoke, Temperature, Pressure sensors & alerts transmission
- Opportunity to prevent accidents through timely actions/alerts

<https://youtu.be/pZsroRmbpYQ>

InsureTech Solutions / Drones

Drones in Industrial/Crop Insurance

- Spherical access photography and image analytics
- Can be used as a pre-underwriting, periodic ongoing surveillance, and post peril data gathering tool

InsureTech Solutions / Computer Vision

Computer Vision in Life Insurance

- Facial analytics technology
- Photo or Video based
- Indicators of Body Mass Index (BMI), Estimated age, Smoking indication
- Clubbed with Dynamic Questioning (use of AI)
- Quick and convenient for low value covers

Low-value life covers see use of facial analytics tech

Mayur.Shetty
@timesgroup.com

Mumbai: Aegon Life Insurance has started using computer vision and facial analytics for underwriting proposals by using a live video to ascertain the age and general well-being of a proposer. The insurer expects to do away with intrusive medical checks for low-value covers using this technology.

Speaking to **TOI**, Aegon Life MD & CEO Satishwar B said that the company has been pilot-testing facial recognition technology to identify gender, age and general well-being. This helps in valida-

ting information already provided by the proposer.

The use of computer vision goes a step beyond teleconsultation, where a medical practitioner engages

The co expects to do away with intrusive medical checks using this technology

with the insurance buyer over a video call.

Aegon has already been using teleconsultation and is now looking at computer vision as a straight-through process. Aegon is part of the Times Group, which publis-

hes this newspaper.

“We started testing with something called an identity check using a 10-15 second video that helped do a face match with the image obtained during the KYC process. We then started testing the same video to identify whether it was male or female, and longevity. We clearly see this as a tech for future,” said Satishwar.

He added the accuracy will improve as the technology is like artificial intelligence and there is constant learning. While currently it only validates the information provided, it could be the only underwriting that is done.

InsureTech Solutions / AI & ML

Claims Handling: The Lemonade example:

- Smartphone app >> generic questions >> damage visuals through smartphone camera.
- AI based analysis >> 18 anti-fraud algorithms.
- Noncomplex claims are approved within seconds >> Complex claims routed to human analysts

Fraud Prevention thru AI : Australian FinTech

- AI scan of social media accounts, criminal records, property and vehicle history and other documentation submitted with the claim

Potential Use Cases of Blockchain Technology

- Health Insurance:
 - Health records updated on Blockchain could be used in underwriting/pricing of a health/life policy
- Smart Contracts:
 - Severity of weather conditions may auto-trigger instant claims pay-outs in Crop Insurance
- P2P insurance:
 - Join >> Contribute >> Share Loss/Gains

Key Chatbot Use cases in Insurance

- Instant quotation
- Claims Filing & Status Enquiry
- Product Features, T&C, Q&A
- Admin changes
- Broker assistance

New Markets

- Slice offers Pay per Use insurance for on-demand workers (e.g Uber Drivers) and home-share insurance for hosts lending out their homes (on Airbnb, FlipKey)
- Cuvva offers hourly insurance if you need to borrow a friend's car
- Metromile offers pay per mile coverage for personal rides
- Trov offers a turn-on, turn-off facility for the cover

New Business Models

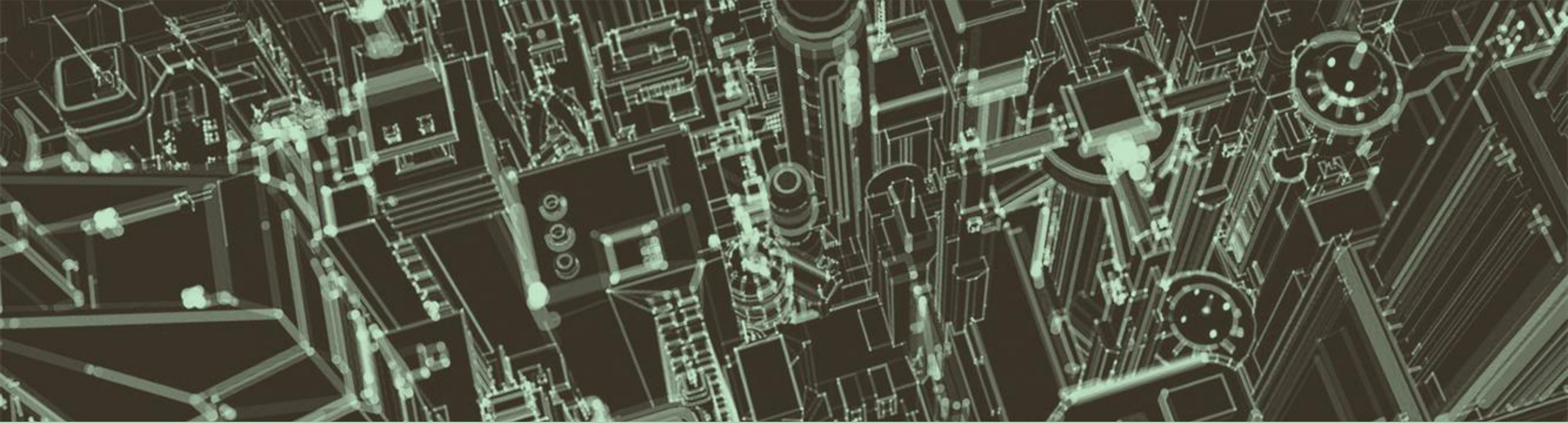
- LAKA charges premiums based on claims experience, while premiums are capped through suitable reinsurance
- Many P2P models redistribute excess profits back to members
- Collaboration between traditional incumbents and insure-techs

A Case Study: UK Pensions

InsureTech examples

- www.metromile.com in Motor Insurance
- www.lemonade.com in General Insurance
- www.policybazaar.com in Insurance Shopping
- www.hioscar.com in Health Insurance
- www.etherisc.com in Blockchain led Insurance (Crop / Flight Delay)

Recap / Questions ?



Thank You

Author & Presenter: Vivek Lakhotia

